# Retail Sales & Inventory Analysis

End-to-end analytical project transforming raw retail data into actionable business intelligence — covering revenue performance, margin analysis, supplier contribution, and inventory risk management.

PYTHON · SQL · TABLEAU

Business Objectives

# What Questions Does This Analysis Answer?

1

### Revenue Drivers

Which products, categories, stores, and suppliers generate the highest sales and gross profit?

2

### Margin Strength

Which product groups deliver stronger or weaker gross margins, and where are the opportunities?

3

### Inventory Risk

Which products are fast-moving, slow-moving, at risk of stockout, or at risk of overstock?

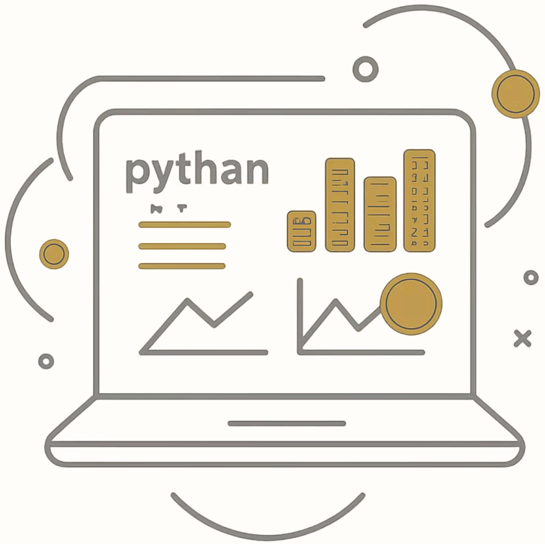
4

### Replenishment

How can inventory and replenishment decisions support better commercial performance?

Tools & Technologies

# The Analytical Stack




The project demonstrates end-to-end data skills — from cleaning and transformation through to KPI development, SQL analysis, and visual storytelling.

### Python

Pandas, NumPy, Jupyter Notebook

### SQL

MySQL for business analysis queries

### Tableau

Interactive dashboard design

Project Workflow

# Three-Stage Analytical Pipeline

**Raw sales and inventory files were cleaned, standardised, and joined in Python, then analysed via SQL for business KPIs, and finally visualised in interactive Tableau dashboards for stakeholder consumption.**

Dashboard Preview

# Business Performance Dashboard

This dashboard provides a comprehensive view of overall business performance, surfacing key sales, profit, and inventory KPIs. It highlights revenue and gross profit trends, product and category performance, supplier contribution, and store-level performance — enabling teams to identify the primary drivers of business growth and profitability at a glance.

Dashboard Preview

# Inventory Risk Analysis Dashboard

This dashboard focuses on inventory health and risk management. It identifies fast-moving and slow-moving products, stockout-risk items, and overstocked inventory — supporting replenishment planning, improving inventory turnover, and reducing carrying costs.


### Fast-Moving Products

13,997 items identified

### Stockout Risk

10,323 products at risk

### Overstock Risk

21,585 products flagged

Key KPIs

# Business Performance at a Glance

## $10.48M

### Net Sales

Total revenue generated across all transactions

## $4.63M

### Gross Profit

Total profit after cost of goods sold

## 44.21%

### Gross Margin

Strong margin indicating healthy pricing power

## 131K

### Transactions

Total customer transactions recorded

## $733K

### Total Returns

~7% of net sales — area for investigation

## $1.10M

### Inventory at Cost

Current stock value requiring optimisation

Key Business Insights

# What the Data Reveals

### Strong Profitability

$10.48M net sales and $4.63M gross profit with a 44.21% margin demonstrate solid commercial performance.

### Returns Warrant Investigation

Returns of $733.57K (~7% of net sales) signal potential product quality, pricing, or fulfilment issues.

### Inventory Imbalance

A mix of stockout-risk and overstocked products highlights the need for smarter replenishment planning.

### Supplier Dependency Risk

Supplier concentration analysis underscores the importance of diversifying procurement for supply chain resilience.

Business Recommendations

# Actionable Next Steps

1. ## Prioritise Replenishment

   Act on stockout-risk products to minimise lost sales opportunities.

2. ## Reduce Overstock

   Review slow-moving inventory to lower carrying costs and improve turnover.

3. ## Investigate Returns

   Analyse high-return products for quality, pricing, or fulfilment issues.

4. ## Monitor Suppliers

   Track supplier performance and concentration to reduce dependency risk.

5. ## Protect Margins

   Evaluate low-margin products before expanding promotional activity.

6. ## Establish KPI Monitoring

   Implement ongoing tracking for sales, profitability, returns, and inventory health.

Skills Demonstrated

# End-to-End Data Analytics Competency

### Data Cleaning & EDA

Transforming raw datasets into analysis-ready inputs using Python and Pandas.

### SQL Business Analysis

Writing complex queries to evaluate revenue, margin, supplier, and inventory metrics.

### Tableau Dashboarding

Designing interactive visualisations for stakeholder decision-making.

### Business Insight Communication

Translating analytical findings into clear, actionable recommendations.

**Repository:** `aditya-pandey-data/retail-sales-inventory-analysis` — includes Jupyter Notebook, SQL scripts, Tableau workbook, and full documentation.